

The Effect of Spaced Repetition on Learning and Knowledge Transfer in a Large Cohort of Practicing Physicians

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Abstract

Purpose

Spaced repetition is superior to repeated study for learning and knowledge retention, but literature on the effect of different spaced repetition strategies is lacking. The authors evaluated the effects of different spaced repetition strategies on long-term knowledge retention and transfer.

Method

This prospective cohort study, conducted from October 1, 2020, to July 20, 2023, used the American Board of Family Medicine Continuous Knowledge Self-Assessment (CKSA) to assess learning and knowledge transfer of diplomates and residents. Participants were randomized to a control group or 1 of 5 spaced repetition conditions during 5 calendar quar-

ters (January 1, 2021, to March 31, 2022). Participants in the spaced repetition groups received 6 repeated questions once or twice. Incorrectly but confidently answered questions were prioritized for repetition, with decreasing priority for questions answered incorrectly with lesser confidence. All participants received 6 rewritten questions corresponding to their initial questions chosen for repetition in quarter 10 (second quarter of calendar year 2023).

Results

A total of 26,258 family physicians or residents who completed the CKSA in the baseline period were randomized. Spaced repetition was superior to no spaced repetition for learning at quarter 6 (58.03% vs 43.20%, $P < .001$,

Cohen $d = 0.62$) and knowledge transfer at quarter 10 (58.33% vs 52.39%, $P < .001$, Cohen $d = 0.26$). Double-spaced repetitions were superior to single-spaced repetitions for learning (62.24% vs 51.83%, $P < .001$, Cohen $d = 0.43$) and transfer (60.08% vs 55.72%, $P < .001$, Cohen $d = 0.20$). There were no meaningful differences in learning or transfer between repetition strategy chosen in the single- or double-repetition groups.

Conclusions

This study affirms the value of spaced repetition in improving learning and retention in medical education and ongoing professional development.

Spaced repetition is an educational technique that uses “the same [specified] material ... distributed over a [specified] number of periods separated by a [specified] interstudy interval, with a [specified] learning outcome after a [specified] retention interval.”¹ Repeated, longitudinal testing over time using spaced repetition results in better long-term retention of learning than repeated study of the same material,^{2–4} particularly

when feedback is provided and answering questions requires application of knowledge through use of short-answer or context-rich multiple-choice questions.^{5,6} Early spaced repetition studies occurred under controlled laboratory conditions with short intervals between repetitions. Most studies have examined a single repetition strategy in small to modestly sized learner cohorts. Spaced repetition improves learning at the undergraduate health professions educational level.^{7–10} Results have been mixed at the graduate medical education level^{11–14} but have shown promise for improving both knowledge^{15,16} and performance^{17–19} among practicing physicians.

Member boards of the American Board of Medical Specialties use longitudinal knowledge assessments in their continuing certification programs, either for lifelong learning and self-assessment or as a means of assessing

cognitive expertise.^{20–22} Although some American Board of Medical Specialties boards have incorporated spaced repetition into their longitudinal assessments,^{23,24} different spaced repetition strategies have not yet been systematically evaluated beyond the short term in large cohorts of physicians under “real-world conditions.” In-training examinations are used longitudinally for assessment in graduate medical education, but they have not as yet incorporated spaced repetition strategies. The goal of this study was to explore, in a large cohort, the effects of different spaced repetition strategies on American Board of Family Medicine (ABFM) diplomate (certified physicians) and family medicine residents’ long-term knowledge retention (learning). We additionally sought to evaluate how these spaced repetition strategies affect transfer of

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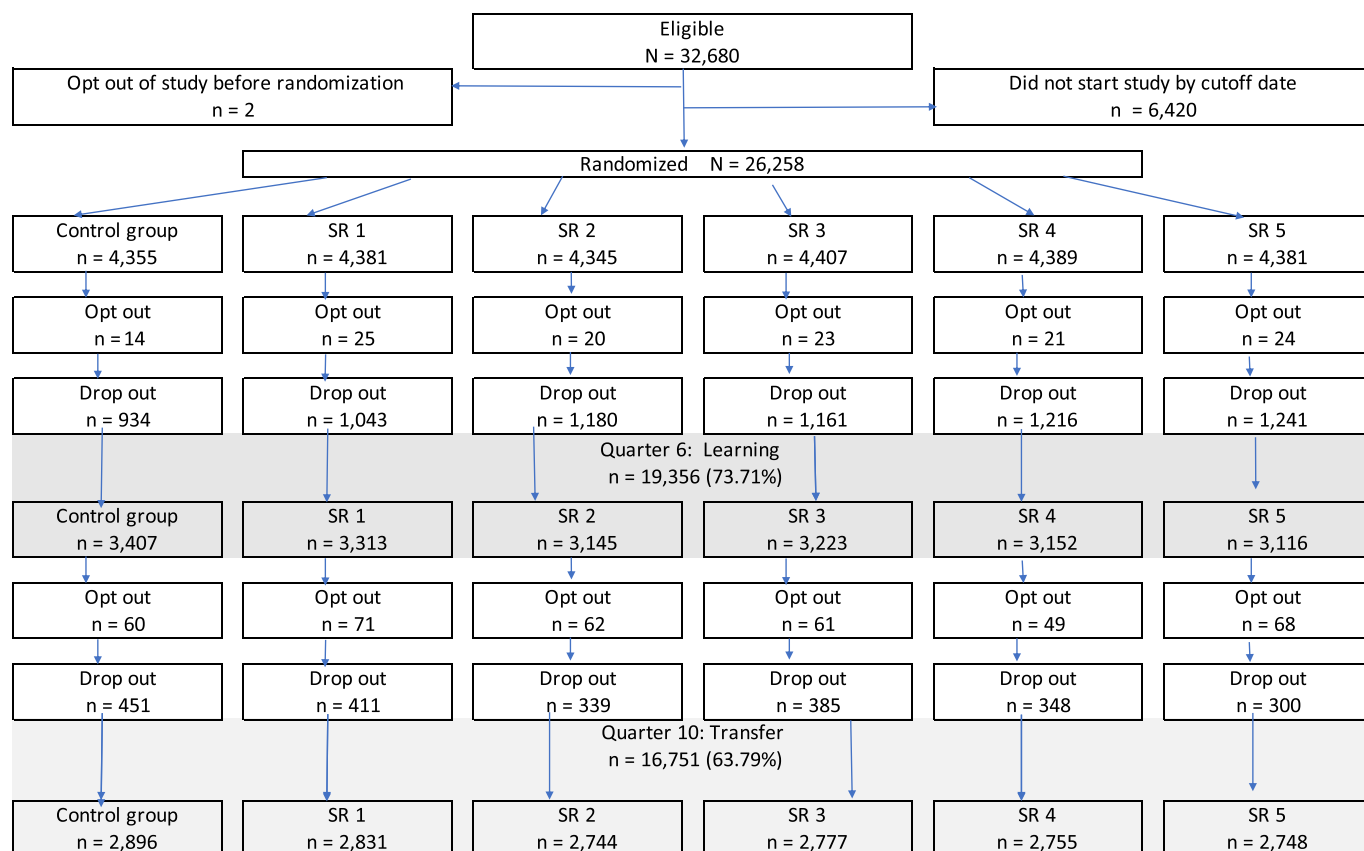


Figure 1 Participant enrollment and randomization in and completion of American Board of Family Medicine spaced repetition (SR) study, 2021 to 2023. Opt out indicates individuals who called and asked to be removed from the study. Dropped out indicates did not complete Continuous Knowledge Self-Assessment (CKSA) questions in a quarter where they were assigned to receive repeat questions, failed to complete CKSA in quarter 6, or failed to complete CKSA in quarter 10.

knowledge from one clinical scenario to another.

Method

Design

This 30-month, prospective, randomized, controlled cohort study, conducted during 10 calendar quarters from October 1, 2020, through July 20, 2023 (allowing a 3-week grace period for completion of questions delivered in the last study quarter), used the Continuous Knowledge Self-Assessment (CKSA) to assess learning and knowledge transfer of participants. The CKSA is one of several options for earning lifelong learning and self-assessment credit in the ABFM's continuing certification program.

Each quarter, 25 test-quality multiple-choice knowledge questions developed by trained item writers were delivered to participants on computer- and/or mobile-enabled platforms. Before submitting answers to each question,

participants rate confidence in their answers on a 6-point scale (from extremely confident to not at all confident). External reference use is permitted. After submitting an answer, participants receive immediate feedback, including answer accuracy, references, and a critique explaining the correct and incorrect answers. There is no time limit per question because the platform is solely formative in nature, with no passing score. Participants receive summary reports indicating their performance across different content areas, which can help them identify knowledge strengths and gaps. After 4 quarters of participation (and quarterly thereafter), participants also receive a prediction of their performance on the ABFM summative knowledge assessment. Currently, more than 30,000 participants begin CKSA questions each quarter. Spaced repetition was not used in the CKSA before the commencement of this study. Ethical approval was granted by the American Academy of Family Physicians Institutional Review Board.

Population

A total of 32,680 U.S. family physicians or family medicine residents who completed the CKSA quarterly questions in the fourth calendar quarter of 2020 were eligible for study inclusion. These individuals were informed about the study and given the opportunity to opt out through the quarterly ABFM newsletter and on signing in to begin the CKSA in the first quarter of 2021. Two individuals declined to participate through the initial opt-out opportunity; an additional 6,420 failed to sign up for the CKSA by the end of the first quarter cutoff date. The remaining 26,258 individuals were included in the study (Figure 1).

Approach and hypotheses

Within the existing quarterly question delivery structure of the CKSA, participants were randomized to a control group or 1 of 5 spaced repetition conditions during the subsequent 5 calendar quarters (January 1, 2021, to

Chart 1
Randomization Groups With Quarterly Repetition Strategies, Learning, and Transfer Assessments in American Board of Family Medicine Spaced Repetition Study, 2021–2023

Cohort	Q4 2020 study quarter 0	Q1 2021 study quarter 1	Q2 2021 study quarter 2	Q3 2021 study quarter 3	Q4 2021 study quarter 4	Q1 2022 study quarter 5	Q2 2022 study quarter 6	Q3 2022 study quarter 7	Q4 2022 study quarter 8	Q1 2023 study quarter 9	Q2 2023 study quarter 10
Control ^a	B						F(EL)				C(T)
SR 1	B	R1					F(EL)				C(T)
SR 2	B		R1				F(EL)				C(T)
SR 3	B	R1	R2				F(EL)				C(T)
SR 4	B	R1		R2			F(EL)				C(T)
SR 5	B		R1		R2		F(EL)				C(T)

Abbreviations: B, baseline (up to 6 questions answered incorrectly chosen as baseline items for repetition); C(T), clinical scenario transfer (delivery of rewritten items to assess transfer of knowledge to a different clinical scenario); F(EL), follow-up of enduring learning (follow-up delivery of baseline items to assess enduring learning); R1, first delivery of baseline items for spaced repetition; R2, second first delivery of baseline items for spaced repetition; SR 1, single-spaced repetition at 1 quarter interval; SR 2, single-spaced repetition at 2 quarter intervals; SR 3, double-spaced repetitions at 1 quarter interval; SR 4, double-spaced repetitions at 1 quarter followed by 2 quarter intervals; SR 5, double-spaced repetitions at 2 quarter intervals.

^aControl group baseline measure of learning that may have occurred from means other than spaced repetition (reading critiques, other learning).

March 31, 2022). We could have tested many different combinations of numbers of repeated questions, numbers of repetitions, and intervals of repeated questions; for operational feasibility, we elected to limit the study to 5 intervention conditions.

Participants continued to receive a total of 25 questions per quarter throughout the study. Participants in the spaced repetition groups received 6 repeated questions once (groups 1–2) or twice (groups 3–5). The single-spaced repetition groups received their questions in quarter 1 (group 1) or quarter 2 (group 2). The double-spaced repetition groups received their repeated questions in quarters 1 and 2 (group 3), quarters 1 and 3 (group 4), or quarters 2 and 4 (group 5; Chart 1). All spaced repetition questions came from the block of 25 CKSA questions in the baseline period and were individualized by participants. Questions answered incorrectly but “extremely confidently” received highest priority for repetition, with decreasing priority for questions answered incorrectly with lesser degrees of confidence. If necessary, correct answers could be selected for spaced repetition, with higher priority for questions answered correctly but not confidently (e.g., guesses) than those answered with higher degrees of confidence (see Supplemental Digital Appendix 1 at <http://links.lww.com/>

ACADMED/B619). Control group participants received no repeated questions during this time. Participants remained in the study unless they notified the ABFM of the desire to opt out or they failed to complete all 25 CKSA questions in the quarter in which they were scheduled to receive repeated questions.

All remaining participants received their 6 repeated questions in quarter 6 of the study (second quarter of calendar year 2022). We defined our first outcome, learning, as the percentage of incorrectly answered baseline questions subsequently answered correctly during quarter 6. This allowed us to compare learning and retention in those groups receiving intervening spaced repeated questions with the control group, who had gone 18 months without seeing repeated questions. We also compared differences in learning and retention between participants receiving 1 repetition compared with those receiving 2, differences between the 2 single-spaced repetition strategies, and differences among the 3 double-spaced repetition strategies. We hypothesized that participants receiving repeated questions would demonstrate better learning at quarter 6 compared with participants not receiving repeated questions, participants receiving double-spaced repetitions would demonstrate better learning at quarter 6 compared with participants

receiving single-spaced repetition, and the interval of spacing within the single- or double-spaced repetitions groups would not result in significant differences in learning at quarter 6.

We further examined the effect of these spaced repetition strategies on subsequent transfer of knowledge from one clinical scenario to another. To assess transfer, we used new rewritten questions,²⁵ emphasizing the same key learning points as the original (baseline) question, framed in different contexts (e.g., different age or gender of patient, variations in presenting symptoms, variations in patient comorbidities, tailoring of first-line treatment options). Each control and intervention group participant received 6 rewritten questions corresponding to their initial questions chosen for repetition in quarter 10 of the study (second quarter of calendar year 2023). We defined transfer as the percentage of rewritten questions from initially incorrect questions answered correctly in quarter 10. We hypothesized that participants in the initial spaced repetition groups would demonstrate better transfer at quarter 10 compared with participants in the control group, participants originally receiving double-spaced repetitions would demonstrate better transfer at quarter 10 compared with participants originally receiving single-spaced repetition, and the interval

Table 1

Spaced Repetition Participant Characteristics at Enrollment, Completion of Learning Phase (Quarter 6), and Completion of Transfer Phase (Quarter 10), American Board of Family Medicine Spaced Repetition Study, 2021–2023^a

Characteristic	All randomized participants (N = 26,258), no. (%)	Participants completing Q6 (learning phase) (n = 19,356), no. (%)	P value for Q6 completers vs noncompleters	Participants completing Q10 (transfer phase) (n = 16,751), no. (%)	P value for Q10 completers vs noncompleters
Age group, y					
< 30	715 (2.72)	367 (1.89)	< .001	239 (1.43)	< .40
30–39	6,034 (22.98)	3,811 (19.69)		3,099 (18.50)	
40–49	6,657 (25.35)	5,039 (26.03)		4,457 (26.61)	
50–59	6,963 (26.52)	5,532 (28.58)		4,973 (29.69)	
60–69	4,862 (18.52)	3,800 (19.63)		3,307 (19.74)	
≥ 70	1,027 (3.91)	807 (4.17)		676 (4.04)	
Gender					
Male	13,299 (50.65)	10,008 (51.70)	< .001	8,764 (52.32)	< .001
Female	12,933 (49.25)	9,327 (48.19)		7,969 (47.57)	
Nonbinary, prefer not to answer, or prefer to self-describe	26 (0.10)	21 (0.11)		18 (0.11)	
Degree					
MD	22,660 (86.30)	16,901 (87.32)	< .001	14,651 (87.46)	< .001
DO	3,598 (13.70)	2,455 (12.68)		2,100 (12.54)	
Graduation status					
International medical graduate	5,710/26,227 (21.77)	3,950/19,333 (20.43)	< .001	3,336/16,732 (19.94)	< .001
U.S. or Canadian medical graduate	20,517/26,227 (78.23)	15,383/19,333 (79.57)		13,396/16,73 (80.06)	
Race					
White	18,136/24,696 (73.44)	13,781/18,364 (75.04)	< .001	12,093/15,999 (75.59)	< .001
Asian	4,105/24,696 (16.62)	2,861/18,364 (15.58)		2,495/15,999 (15.59)	
Black or African American	1,328/24,696 (5.38)	934/18,364 (5.09)		769/15,999 (4.81)	
American Indian or Alaska Native	185/24,696 (0.77)	143/18,364 (0.78)		120/15,999 (0.75)	
Native Hawaiian or other Pacific Islander	116/24,696 (0.47)	85/18,364 (0.46)		73/15,999 (0.46)	
Other	826/24,696 (3.34)	560/18,364 (3.05)		449/15,999 (2.81)	
Ethnicity					
Not Hispanic or Latino	23,110/24,696 (93.58)	17,292/18,364 (94.16)	< .001	15,093/15,999 (94.34)	< .001
Hispanic or Latino	1,586/24,696 (6.42)	1,072/18,364 (5.84)		906/15,999 (5.66)	
Time since initial certification, y					
Resident	241/24,850 (0.97)	71/18,565 (0.38)	< .001	44/16,233 (0.27)	< .001
< 5	3,571/24,850 (14.37)	2,089/18,565 (11.25)		1,686/16,233 (10.39)	
5–10	4,285/24,850 (17.24)	3,169/18,565 (17.07)		2,723/16,233 (16.77)	
11–20	6,996/24,850 (28.15)	5,485/18,565 (29.54)		4,914/16,233 (30.27)	
21–30	5,936/24,850 (23.89)	4,754/18,565 (25.61)		4,271/16,233 (26.31)	
31–40	3,192/24,850 (12.85)	2,501/18,565 (13.47)		2,166/16,233 (13.34)	
> 40	629/24,850 (2.53)	496/18,565 (2.67)		429/16,233 (2.64)	

(Table continues)

Table 1

(Continued)

Characteristic	All randomized participants (N = 26,258), no. (%)	Participants completing Q6 (learning phase) (n = 19,356), no. (%)	P value for Q6 completers vs noncompleters	Participants completing Q10 (transfer phase) (n = 16,751), no. (%)	P value for Q10 completers vs noncompleters
Most recent FMCE score					
< 400	1,144/26,104 (4.38)	780/19,304 (4.04)	< .001	669/16,722 (4.00)	< .001
400–500	9,513/26,104 (36.44)	6,754/19,304 (34.99)		5,778/16,722 (34.55)	
501–600	9,493/26,104 (36.37)	7,038/19,304 (36.46)		6,067/16,722 (36.28)	
601–700	4,644/26,104 (17.79)	3,657/19,304 (18.94)		3,226/16,722 (19.29)	
> 700	1,310/26,104 (5.02)	1,075/19,304 (5.57)		982/16,722 (5.87)	
Direct patient care					
Yes	18,085/19,852 (91.10)	14,189/15,588 (91.03)	.77	12,565/13,778 (91.20)	.30
No	1,767/19,852 (8.90)	1,399/15,588 (8.97)		1,213/13,778 (8.80)	
Faculty					
Core or salaried faculty	1,349/17,173 (7.86)	1,079/13,453 (8.02)	.20	948/11,919 (7.95)	.71
Volunteer or clinical faculty	2,411/17,173 (14.04)	1,911/13,453 (14.21)		1,687/11,919 (14.15)	
No	13,413/17,173 (78.11)	10,463/13,453 (77.77)		9,284/11,919 (77.89)	
Practice size					
Solo practice	1,667/19,969 (8.35)	1,265/15,587 (8.12)	.11	1,112/13,784 (8.07)	.03
2–5 Practitioners	6,480/19,969 (32.45)	5,028/15,587 (32.26)		4,416/13,784 (32.04)	
6–20 Practitioners	6,649/19,969 (33.30)	5,251/15,587 (33.69)		4,675/13,784 (33.92)	
> 20 Practitioners	5,173/19,969 (25.91)	4,043/15,587 (25.94)		3,581/13,784 (25.98)	
Practice rurality					
Urban	14,089/16,753 (84.10)	10,422/12,426 (83.87)	.67	9,093/10,849 (83.81)	.60
Large rural	1,460/16,753 (8.71)	1,098/12,426 (8.84)		962/10,849 (8.87)	
Small rural	868/16,753 (5.18)	648/12,426 (5.21)		566/10,849 (5.22)	
Isolated	336/16,753 (2.01)	258/12,426 (2.08)		228/10,849 (2.10)	

Abbreviations: FMCE, Family Medicine Certification Examination; IMG, international medical graduate; Q6, quarter 6; Q10, quarter 10.

^aCategory values not totaling to overall values indicate missing data.

of spacing within the original single- or double-spaced repetitions groups would not result in significantly different transfer at quarter 10.

Statistical analysis

We used χ^2 independence tests to identify differences in selected demographic and practice variables between individuals remaining in the study and those who dropped out by the end of quarter 6 and at quarter 10. To evaluate learning, we used individuals' correct percentage as the main outcome variable. We used t tests to compare the effects of spaced repetition vs no spaced repetition and the effects of 2 repetitions vs 1. We also used t tests to compare the correct percentages between the single-spaced repetition groups and

1-way ANOVA with the correct percentages between the 2 repetition strategies. To control for the potential of inflated type 1 error introduced by large sample size, we also calculated the effect size (Cohen d for t test^{26,27} and η^2 for ANOVA²⁸). We used R, version 4.0.2 (R Foundation for Statistical Computing, Vienna, Austria) to perform all statistical analyses.

Results

Participants

A total of 26,258 participants were sequentially randomized on logging into the CKSA to the control group or 1 of the 5 spaced repetition conditions. Each study condition contained both

diplomates and residents. A total of 19,356 individuals (73.71% of starters) completed the study through quarter 6, where learning was assessed. Of these individuals, 16,751 (86.5% of quarter 6 participants and 63.79% of starters) completed the CKSA in the second quarter of 2023, when transfer was assessed. Figure 1 depicts participant enrollment, allocation, and completion by stage. Demographics of the initial study group, quarter 6, and quarter 10 completers are given in Table 1. Compared with those who dropped out of the study, completers at both quarter 6 and quarter 10 were statistically more likely to be male, to be White and not Hispanic or Latino, to be between the ages of 40 and 69 years, and to have an

Table 2

Spaced Repetition Participant Learning (Quarter 6) and Transfer (Quarter 10) Outcomes by Number of Spaced Repetitions, American Board of Family Medicine Spaced Repetition Study, 2021–2023

Group	Learning, mean (SD), % of correct responses	t Test result	Cohen d (range)
Quarter 6 (learning)			
Control (n = 3,407)	43.20 (23.99)	t = 30.43, df = 4246.3, P < .001	0.62 (0.58–0.66)
Any SR (n = 15,949)	58.03 (24.93)		
1 SR groups (n = 6,458)	51.83 (24.76)	t = 24.54, df = 11731, P < .001	0.43 (0.39–0.46)
2 SR groups (n = 9,491)	62.24 (24.16)		
Quarter 10 (transfer)			
Control (n = 2,896)	52.39 (22.84)	t = 12.80, df = 4229.2, P < .001	0.26 (0.22–0.30)
Any SR (n = 13,855)	58.33 (23.15)		
1 SR groups (n = 5,575)	55.72 (23.17)	t = 11.72, df = 11,826, P < .001	0.20 (0.17–0.24)
2 SR groups (n = 8,280)	60.08 (22.97)		

Abbreviation: SR, spaced repetition.

allopathic degree from a medical school in the United States or Canada. They were also more likely to be in the middle of their career and to have scored greater than 600 on their most recent certification examination. These statistical differences likely result from the large sample size; because most between-group differences were less than 2%, they are likely of minimal real-world consequence.

Questions

All participants in the repetition cohorts received 6 repeated questions in their designated quarter. All participants received 6 repeated questions in quarter 6 (evaluation of learning) and 6 rewritten questions in quarter 10 (evaluation of transfer). Most responses chosen for spaced repetition were initially answered incorrectly; less than 1% of the repeated questions were initially answered correctly (with low confidence) but later answered incorrectly (e.g., forgotten). Because of the small number of possibly forgotten repeated questions and because most participants received repeats of only baseline incorrect questions, we could not determine between-group differences in forgetting at the individual participant level. We therefore limited our analyses to repetitions of questions initially answered incorrectly. Additionally, 1 question was removed from analysis after deployment due to participant comments about evidence and concerning wording.

Results were calculated on a median of 6 repeated (for learning) or rewritten (for transfer) questions per group assignment (means, 5.41–5.47; SDs, 0.82–0.91; see Supplemental Digital Appendix 2 at <http://links.lww.com/ACADMED/B619>).

Learning

Repetition resulted in greater learning. At the end of quarter 6, 43.20% of repeated initially incorrect questions were

answered correctly by control group participants. Overall, spaced repetition resulted in more learning than no spaced repetition, with a medium to large effect size (58.03% vs 43.20%, $P < .001$, Cohen $d = 0.62$ [range, 0.58–0.66]). Double-spaced repetitions resulted in more learning (62.24%) than single-spaced repetition (51.83%), with a small to medium effect size ($P < .001$, Cohen $d = 0.43$ [range, 0.39–0.46]; Table 2). Differences between the single-spaced repetition groups were not significant (52.13% vs 51.55%, $P = .10$). In the double-spaced repetition groups, there were small differences favoring the 2-quarter strategy (63.63% vs 61.99% vs 61.14%, $P < .001$), but the effect size ($\eta^2 = 0.001$) was very small (Table 3).

Transfer

Table 2 indicates that the original groups receiving spaced repetition more successfully answered transfer questions correctly than the original control group (58.33% vs 52.39%, $P < .001$, Cohen $d = 0.26$ [range, 0.22–0.30], small to medium effect size). Participants in the original double-spaced repetition groups more successfully answered transfer questions correctly than those in the original single-spaced repetition group (60.08% vs 55.72%, $P < .001$, Cohen $d = 0.20$ [range, 0.17–0.24], small effect size). Differences between the 2 original single-spaced repetition groups (55.85% vs

Table 3

Differences in Learning (Quarter 6) and Transfer (Quarter 10) Among Spaced Repetition Participants in 1 and 2 Spaced Repetition Groups, American Board of Family Medicine Spaced Repetition Study, 2021–2023

Group	Learning, mean (SD), % of questions answered correctly	t Test and 1-way ANOVA	Effect size
Quarter 6 (learning)			
SR 1 (n = 3,313)	51.55 (25.04)	$t = 1.6556, df = 5572.9,$ $P = .10$	NA
SR 2 (n = 3,145)	52.13 (24.47)		
SR 3 (n = 3,223)	61.14 (23.95)	$F = 5.819, df = 2, 8,277,$ $P < .001$	$\eta^2 = 0.001$
SR 4 (n = 3,152)	61.99 (24.19)		
SR 5 (n = 3,116)	63.63 (24.29)		
Quarter 10 (transfer)			
SR 1 (n = 2,831)	55.85 (23.21)	$t = 0.75007, df = 5568.4,$ $P = .45$	NA
SR 2 (n = 2,744)	55.60 (23.13)		
SR 3 (n = 2,777)	60.18 (22.95)	$F = 1.192, df = 2, 8,277,$ $P = .30$	NA
SR 4 (n = 2,755)	60.36 (23.02)		
SR 5 (n = 2,748)	59.71 (22.93)		

Abbreviations: NA, not applicable; SR, spaced repetition.

55.60%, $P = .45$) and between the 2 original double-spaced repetition groups (59.71% vs 60.36% vs 60.18%, $P = .30$) were not significant (Table 3).

Discussion

In this largest and longest randomized study of different repetition strategies in quarterly longitudinal assessment, we found that spaced repetition of case-based multiple-choice questions increased learning at 18 months compared with no spaced repetition. Double-spaced repetition strategies were more effective than single-spaced repetition strategies, with an approximate gain of 8% to 10% per repetition. We also found a smaller but positive effect of spaced repetition for transfer of knowledge from one case scenario to another, with an approximate gain of 3% to 5% per repetition. For both learning and transfer, the repetition strategies within the single- or double-spaced repetition cohorts did not result in important differences.

Interestingly, even without spaced repetition, we found a substantial amount of learning occurred in the control group, which did not receive any spaced repetition in the 18-month period. The control group was intended to account for learning that occurred from immediate feedback (critiques and references) after first answering the question or subsequent self-directed or workplace learning. Given the long duration between question exposure, it seems less likely that the 43% learning rate in the control group was substantially due to question recognition.

We calculated effect sizes in our analyses to gauge the magnitude of differences found. Our findings have important implications not only for board certification but also for health professions education. Our study is consistent with other studies showing that spaced repetition increases knowledge retention.^{2–6,12,15,16} In the educational literature, a Cohen d of 0.2 to 0.25 is often considered a minimally significant effect size.²⁹ The learning effect size of 0.62 in our study is consistent with a Cohen d of approximately 0.5 found in many positive educational interventions,³⁰ suggesting a practical real-world difference. The

differences we found between numbers of spaced repetitions are in line with a mean 13% difference between study groups found in an earlier study.⁶ There were no substantial differences between the single-spaced repetition or the double-spaced repetition strategies evaluated. This finding is consistent with studies suggesting that the number of repetitions is more important than the timing of repetitions.^{31,32} Our study also supports recommendations that repetitions occur at widely spaced intervals.³³

Our findings are also consistent with previous studies suggesting that spaced repetition is effective for transfer.^{10,17–19,33–35} Several factors might account for the smaller effect size with transfer than with learning. This finding is unlikely due to additional loss of participants between quarters 6 and 10 because the sample size for the transfer analysis was still substantial and the difference between the upper and lower bounds of effect size CIs is similar in both the learning and transfer analyses. Given the large number of participants in each cohort and the randomized nature of this study, we think it unlikely that more interval learning (outside the study) or more learning from reading the educational critiques or references occurred in the control group than the other groups. Even though retrieval practice facilitates knowledge transfer,³³ a smaller transfer effect is consistent with findings from a recent meta-analysis.³⁴ We posit that additional cognitive effort or reflection may be needed to transfer learning to a different scenario.

The design of our study might also partially explain this finding. The study was designed primarily to assess learning. At quarter 6, when learning was assessed, all groups received another repeat of their baseline questions (all selected from the 25 CKSA questions delivered in 2020 quarter 4). Therefore, the quarter 10 results essentially represent transfer based on 1, 2, or 3 previous repetitions. The relative leveling at approximately 60% correctness (see Supplemental Digital Appendix 1 at <http://links.lww.com/ACADMED/B619>) between 2 spaced repetitions for learning at quarter 6 and 3 spaced repetitions at quarter 10 for transfer aligns with previous studies suggesting that 3 repetitions may provide the best balance of improved learning

with decreasing improvements with subsequent repetitions.^{30,31} To assess differences between no previous repetitions and transfer, we would have needed an additional cohort receiving only the baseline questions before the transfer question. A true 0 repetition group might perform somewhere between the 43% correct rate seen in the control group learning quarter 6 and the 52% correct rate at quarter 10, which would potentially increase the effect size.

It is possible that the 6% difference (no repetition vs any repetition) or 8% difference (no repetition vs 2 repetitions) for transfer, continued over time, covering a wide array of content in large numbers of physicians caring for large populations of patients, could correlate with real-world differences in patient outcomes. Further refinement and targeting of spaced repetition strategies (such as tailoring to specific physicians with larger knowledge or performance gaps) might also increase the eventual effectiveness of spaced repetition.

We could have examined additional numbers and intervals of repetitions or selected a higher number of questions for repetition. However, we needed to balance the operational and technical complexity of additional strategies with the need to minimize potential disruption in an already functioning, popular, and highly successful longitudinal assessment platform. The CKSA platform is currently fixed at 25 questions per quarter—adding more repeated questions would have resulted in fewer new questions on different content areas. The CKSA participants value the platform for its coverage of the broad scope of family medicine. Allowing more than 6 repeated questions would have affected the ability to deliver estimates of summative examination performance after 4 quarters of CKSA participation. In the future, we could potentially modify the platform to deliver more questions per quarter to allow for more spaced repetitions while still providing content coverage. These modifications were not feasible during the time of the study.

Strengths of our study include large numbers of participants, concurrent examination of different number and spacing repetition strategies, and the extended period for outcomes assessment (18 months for learning, 30 months for

transfer). Several limitations should be considered. First, differential use of references to answer questions between study groups could affect answer accuracy at the 18- and/or 30-month outcomes assessments. With randomization across cohorts and large cohort sizes, we do not expect that this would significantly affect the results. Second, although repeated questions were deployed on a specific quarterly schedule based on cohort, participants were able to complete their questions any time within a given quarter. Thus, the actual spacing between answering of repeated questions was likely more variable in our study than in the more controlled and narrower timeframes in laboratory or classroom studies. With randomization, large cohort sizes, and previous studies suggesting the repetition interval is less important than the number of repetitions,^{30,31} we do not suspect significant differences between groups that would bias the outcomes. Third, confidence ratings are highly subjective, likely differing between participants and even within participants over time. This would be more concerning if confidence was factored into summative judgments of knowledge (i.e., used in part to determine passing scores). We feel that confidence variability between and within participants is less important when confidence is used solely to stimulate reflection and provide feedback for further education. As above, it is also unlikely that significant group-level differences occurred to the extent of biasing outcomes. Fourth, although written with similar learning points from the initial question, high-quality rewritten clinical scenario-based questions likely take more time to develop than isomorphic (simple or minor) changes often used to develop cloned questions.^{25,36} This study was not designed to compare the effectiveness of more intensive rewritten questions with simpler question modifications on transfer. Fifth, our study was limited to physicians of a single specialty. Although the sound foundation of spaced repetition in the adult learning literature suggests that the findings should be similar in other specialties and other health care professionals, other studies could be considered to verify and quantify these findings. Although more research is needed, a recent systematic review³⁷ of primarily short-term spaced repetition studies suggests that our findings could apply to different medical specialties, different medical professionals, and different countries.

Spaced repetition may affect learning and/or transfer differently in different cohorts of physicians based on characteristics (e.g., age, time since training, initial certification examination score) or practice arrangement (e.g., size, rurality, arrangement, academic affiliation) characteristics. Identifying differences could inform strategies to tailor spaced repetition strategies to different cohorts of physicians. A previous study³⁸ of undergraduate students suggests that spaced repetition increases retention of initially low-confidence responses. Additional analyses could more specifically evaluate this phenomenon in practicing physicians. Longitudinally, studies could examine whether spaced repetition improves metacognitive accuracy (accuracy and confidence alignment) of practicing physicians. Building on earlier work,^{6,11–18} studies on the effect of spaced repetition on practice could include changes in participant diagnostic accuracy over time, evaluating how participants report implementing spaced repetition learning in practice, and additional studies examining the effect and magnitude of spaced repetition with quality parameters, sustainability of initial practice changes,¹⁸ and patient outcomes over time. Several other areas are ripe for exploration. In this study, we targeted questions answered incorrectly but confidently for repetition; other strategies for question selection, such as selecting content based on performance gaps, important new evidence, or high-priority health and safety issues could be explored.

Conclusions

The findings in a real-world certification cohort raise important considerations for those who design formative assessments and other educational programs. We discovered, at least in this prospective, randomized cohort of practicing family physicians and family medicine residents, that, although the pattern of repetitions is less critical, the number of repetitions may play a significant role in learning, retention, and potentially transfer. Furthermore, the study highlights the potential for tailoring spaced repetition strategies to different cohorts of physicians based on characteristics such as age, initial certification examination score (as a proxy of baseline knowledge), and practice arrangement. Further research is needed before extrapolating findings from a single specialty as well as

to assess its effect on diagnostic accuracy, implementation in practice, and patient outcomes over time. Overall, the findings underscore and affirm the value of spaced repetition in improving learning and retention in medical education and ongoing professional development.

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